

Lec. 06.

MATRIX INVERSION

Determinant of a matrix
Exists only for square matrices

For a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$|A| = ad - bc$$

$\det(A)$
Determinant
of A

$$\text{Ex: } \begin{bmatrix} 2 & -3 \\ 1 & -2 \end{bmatrix}$$

$$\begin{aligned} \det(A) &= 2(-2) - (1)(-3) \\ &= -4 + 3 = -1 \end{aligned}$$

For a larger matrix Determinant
can be evaluated as follows

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{bmatrix}$$

$$|A| = \sum_{j=1}^m a_{ij} c_{ij}$$

c_{ij} = cofactor of a_{ij}

$$c_{ij} = (-1)^{i+j} m_{ij}$$

minor ij

Determinant of the reduced matrix obtained by removing row i & column j .

Example:

$$A = \begin{bmatrix} 7 & 2 & 1 \\ 0 & 3 & -1 \\ -3 & 4 & -2 \end{bmatrix}$$

3x3 ←

$$|A| = ?$$

Let us expand it along 1st row

$$|A| = 7C_{11} + 2C_{12} + 1C_{13}$$

$$C_{11} = (-1)^{1+1} M_{11}$$

$$= (-1)^2 \begin{vmatrix} 3 & -1 \\ 4 & -2 \end{vmatrix}$$

$$= -2$$

Determinant of reduced matrix after removing 1st row & 1st column.

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 0 & -1 \\ -3 & -2 \end{vmatrix}$$

$$= (-1)(-3) = +3.$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 0 & 3 \\ -3 & -4 \end{vmatrix}$$

$$= +9$$

$$\begin{aligned}
 \therefore \det(A) &= 7C_{11} + 2C_{12} + C_{13} \\
 &= 7(-2) + 2 \times 3 + 9 \\
 &= -14 + 6 + 9 \\
 &= 1.
 \end{aligned}$$

$$\boxed{\therefore |A| = 1}$$

INVERSE OF MATRIX

Defined for an $n \times n$
square matrix.

Let A be an $n \times n$ matrix
Inverse of A is denoted by A^{-1}
satisfies

$$A \cdot A^{-1} = A^{-1} \cdot A = I.$$

If inverse exists, A is termed
INVERTIBLE

if inverse does not exist,
matrix is **SINGULAR**.

$n \times n$ matrix is invertible
if $|A| \neq 0$.

$$A^{-1} = \frac{\text{adj}(A)}{|A|}$$

adjoint of A

$$= C^T$$

C = matrix of
Cofactors

C is
obtained from
 A by replacing
each element
with its cofactor.

$\nearrow$

$$A = \begin{bmatrix} 7 & 2 & 1 \\ 0 & 3 & -1 \\ -3 & 4 & -2 \end{bmatrix}$$

3×3

$$|A| = 1.$$

matrix of cofactors

$$C = \begin{bmatrix} -2 & 3 & 9 \\ 8 & -11 & -34 \\ -5 & 7 & 21 \end{bmatrix}$$

$$\begin{aligned} & \text{cofactor } 3,2 \\ &= (-1)^{3+2} M_{3,2} \end{aligned}$$

$$\text{adj}(A) = C^T$$

$$= \begin{bmatrix} -2 & 8 & -5 \\ 3 & -11 & 7 \\ 9 & -34 & 21 \end{bmatrix}$$

$$|A| = 1$$

$$A^{-1} = \frac{1}{|A|} \text{adj}(A)$$

$$= \frac{1}{1} \begin{bmatrix} -2 & 8 & -5 \\ 3 & -11 & 7 \\ 9 & -34 & 21 \end{bmatrix}$$

inverse of matrix A.